Type MS · Judge whether data support a claim and spot their limits

Answer 1 of 8 · Question 7 part c.iii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) (i) | C; (ureter)<br><br>*A is a blood vessel*<br>*B is a blood vessel*<br>*D is not attached to kidney* | | 1 |
| (ii) | A; (bladder)<br><br>*B is part of the reproductive system*<br>*C is part of the reproductive system*<br>*D leads from the bladder* | | 1 |
| (b) | • more oxygen;<br>• less carbon dioxide;<br>• more urea; | accept reverse for blood leaving kidney | 3 |
| (c) (i) | 16.8 - 3.6 = 13.2;<br>13.2 ÷ 3.6 =3.66;<br>3.66 × 100 = 366.7%; | | 3 |
| (ii) | • urea;<br>• large/significant increase in protein-rich diet; | | 2 |
| (iii) | Any four from<br><br>• urea, uric acid and ammonia masses are different between the two diets;<br>• ref. creatinine;<br>• amount same/nearly same;<br>• in both diets;<br>• possible result of muscle breakdown; | | 4 |

Total 14 marks

## Type MS · Judge whether data support a claim and spot their limits

### Answer 1 of 8 · Question 7 part c.iii · 4 marks

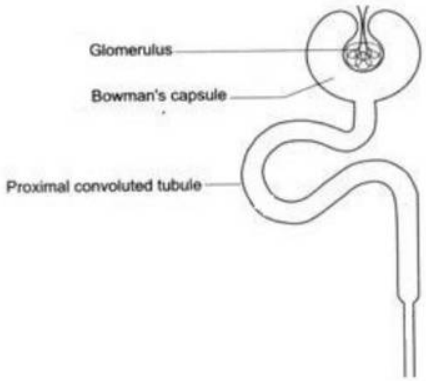
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.iii in this mark scheme extract

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Type MS · Judge whether data support a claim and spot their limits

Answer 2 of 8 · Question 9 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 9 (a) (i) | • evidence of 23500 – 4000 (= 19500) (1)<br>• 19500 ÷ 4000 (x 100)(1)<br>• 488(%) | Allow full marks for correct final answer. Allow 487.5(%)<br><br>Allow ecf if incorrect answer from mp1 – max 2 marks | 3 |
| (ii) | other lifestyle factors/genetic predisposition could have an influence/named lifestyle factor(1) | Allow gender/age | 1 |
| (b) | A diagram similar to:<br>  <br> | One mark for diagram. One mark for each correct label. | 4 |
| (c) | any four from<br>• ultrafiltration does not take place(1)<br>• in glomerulus(1)<br>• less/no waste products/named waste product/water/glucose pass through to tubule(1)<br>• greater volume of waste in blood/more water in blood(1)<br>• glucose/ions not reabsorbed(1)<br>• glucose found in urine(1)<br><br>OR<br><br>any four from<br><br>• blood not filtered effectively/ultrafiltration ineffective(1)<br>• in glomerulus (1) | | 4 |

Type MS · Judge whether data support a claim and spot their limits

Answer 2 of 8 · Question 9 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| | • larger molecules/named larger molecules pass through to tubule/Bowmans capsule(1)<br>• not all glucose/ions reabsorbed(1)<br>• glucose/protein found in urine/urine more concentrated(1) | | |

**Total for Question 9 = 12 marks**

Type MS · Judge whether data support a claim and spot their limits

Answer 3 of 8 · Question 7 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | • axes labelled with units;<br>• correct way round and suitable scale;<br>• correct plots;<br>• suitable line drawn; | | 4 |
| | (ii) | • only not working correctly when rate falls below 100;<br>• data/year not shown when that occurs/already below 100; | | 2 |
| | (iii) | • 2021/2022;<br>• rate falling at 5-10 units per annum/similar values each year; | | 2 |
| | (iv) | • not very accurate/results vary/fluctuate;<br>• earlier data showed improvement in function;<br>• could happen again/results may change; | | 3 |
| (b) | | any three from<br>• protein broken down into urea/deamination;<br>• toxic;<br>• needs to be excreted;<br>• incorrectly functioning kidney can't remove it all if there is a lot;<br>• less protein less urea; | | 3 |

Total 14

## Type MS · Judge whether data support a claim and spot their limits

### Answer 3 of 8 · Question 7 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

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Type MS · Judge whether data support a claim and spot their limits

Answer 4 of 8 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | Increase/goes up; | allow equivalent | 1 |
| | | (ii) | Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;; | Allow one mark for reference to lifestyles only | 2 |
| | (b) | | Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed;<br>Increased risk of (lung/respiratory) infection/coughing; | | 1<br>1 |
| | (c) | (i) | <table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table> | Allow for SA:V ratio 6:3 ECF | 3 |
| | | (ii) | Smaller surface area to volume ratio (in lungs with emphysema);<br>Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide;<br>Less oxygenated blood/less oxygen transferred (to cells);<br>Less (aerobic) respiration; | | Max 3 |

## Type MS · Judge whether data support a claim and spot their limits

### Answer 4 of 8 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Judge whether data support a claim and spot their limits

Answer 5 of 8 · Question 7 part a.iii · 2 marks

2020 JAN · Paper 01 · find part a.iii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 7 | (a) | (i) | • 67.5 ÷ 48.3 (1)<br>• 1:1.4 (1) | Full marks for correct final answer | 2 |
| | | (ii) | Any two from<br>• more males than females are overweight in all countries (1)<br>• lowest percentage of people between the ages of 18-24 in all countries are overweight (1)<br>• increased in all countries 18 - 64 (1) | | 2 |
| | | (iii) | sample size too small (1)<br>not all overweight people identified/sample not representative (1) | | 2 |
| | (b) | (i) | Any three from:<br>• pancreas (1)<br>• releases insulin (1)<br>• reference to the liver taking up glucose (1)<br>• glucose converted to glycogen (1) | R sugar<br>R breaks down glucose | 3 |
| | | (ii) | Any two from:<br>• high blood glucose increases the concentration of the blood (1)<br>• water moves out of cells (1)<br>• by osmosis (1)<br>• excess water in blood excreted in urine (1) | | 2 |

**Total for Question 7 = 11 marks**

Type MS · Judge whether data support a claim and spot their limits

Answer 6 of 8 · Question 3 part b.iii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | A (cornea and lens);<br><br>B is incorrect as the iris does not play a role in focusing light<br>C is incorrect as the iris does not play a role in focusing light<br>D is incorrect as the pupil does not play a role in focusing light | | 1 |
| | (b) | (i) | Points as follows:<br>150:2.7<br>200:3.2<br>250:2.3; | | 1 |
| Clip (b)(i) with (b)(ii) and (b)(iii) | | (ii) | Line of best fit missing out point at 150:2.7 | Ignore lines that extended beyond points | 1 |
| | | (iii) | 150 lux/150:2.7/2.7 diameter; | | 1 |
| | | | does not follow the pattern of other points/trend/not on the line of best fit; | | 1 |
| | | (iv) | As light intensity increases diameter decreases; | ORA | 1 |
| | (c) | | Idea that distant/far away objects are blurred/difficult to see/ORA;<br>as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much; | Reject can't see | 1<br>1 |

**Total question 3 = 8 marks**

Type MS · Judge whether data support a claim and spot their limits

Answer 6 of 8 · Question 3 part b.iii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Judge whether data support a claim and spot their limits

Answer 7 of 8 · Question 7 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | • axes labelled with units;<br>• correct way round and suitable scale;<br>• correct plots;<br>• suitable line drawn; | | 4 |
| | (ii) | • only not working correctly when rate falls below 100;<br>• data/year not shown when that occurs/already below 100; | | 2 |
| | (iii) | • 2021/2022;<br>• rate falling at 5-10 units per annum/similar values each year; | | 2 |
| | (iv) | • not very accurate/results vary/fluctuate;<br>• earlier data showed improvement in function;<br>• could happen again/results may change; | | 3 |
| (b) | | any three from<br>• protein broken down into urea/deamination;<br>• toxic;<br>• needs to be excreted;<br>• incorrectly functioning kidney can't remove it all if there is a lot;<br>• less protein less urea; | | 3 |

Total 14

## Type MS · Judge whether data support a claim and spot their limits

### Answer 7 of 8 · Question 7 part a.ii · 2 marks

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Type MS · Judge whether data support a claim and spot their limits

Answer 8 of 8 · Question 6 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 (a) (i) | 2500 males;<br>$\frac{2500 \times 0.8}{100}$;<br>= 20; | | ecf if use 5000 as starting figure<br>40 scores 2 marks<br>any number×0.8=1 | 3 |
| (ii) | small number/ not enough tested; | | | 1 |
| (iii) | • many more males than females with condition;<br>• show that it is sex-linked;<br>• to X chromosome;<br>• females are XX /males one X chromosome/XY<br>• females can be carriers( so fewer)/males cannot be carriers( so more);<br>• (relatively) small number (in whole population)/small chance of being colour blind;<br>• indicates that it is recessive; | | | 5 |
| (b) | parents | $X^BY \times X^BX^b$; | | 4 |
| | gametes | $X^B$ Y $X^B$ $X^b$; | | |
| | offspring genotypes | $X^B X^B$ $X^BX^b$ $X^BY$ $X^bY$; | | |
| | offspring phenotypes | normal carrier/ disease normal red-green; | | |
| | | female female male male | If parents are incorrect ecf for mp 2,3 and 4<br><br>If X and Y are not used- max of 1 mark | |
| | | | | **Total = 13** |

## Type MS · Judge whether data support a claim and spot their limits

### Answer 8 of 8 · Question 6 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |